

Interferometric Measurement of the Spectral Width of Pulsed Optical Signals

Riley Hebert, Brian J. Smith, Smith Lab Group

Oregon Center for Optical, Material, and Quantum Science and Department of Physics, University of Oregon, Eugene, OR (USA)

Motivation

Classically, we can use optical pulses to achieve light speed communication. Doing so requires receiving custom pulses, and characterizing them requires knowing their spectral width. However, a pulse with duration on the order of picoseconds has a bandwidth too narrow to resolve with a spectrometer. This experiment aims to use Fourier-transform spectroscopy via a Michelson interferometer instead to measure the spectral width.

Theory

A pulse is made of many different frequencies.

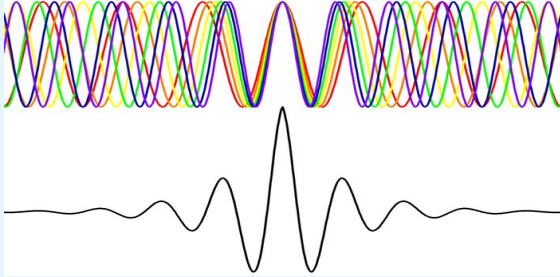


Image source: PennState (<https://www.e-education.psu.edu/mcl-optpro/theory/node795>)

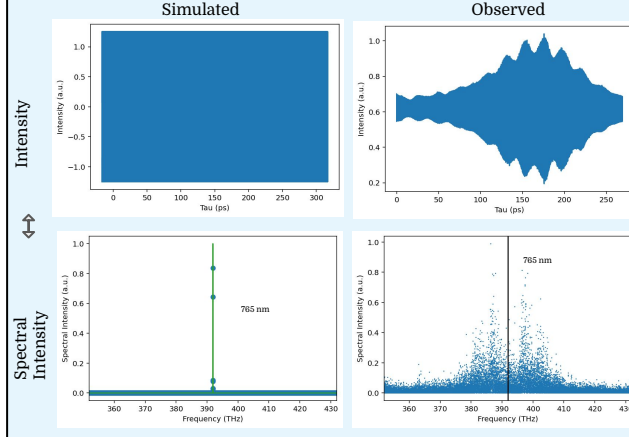
Intensity of interfering identical beams as function of their relative time shift τ :

$$I(\tau) = |A_0|^2 \sigma \sqrt{\frac{\pi}{2}} e^{-\frac{\tau^2}{2\sigma^2}} \cos(2\pi f_0 \tau)$$

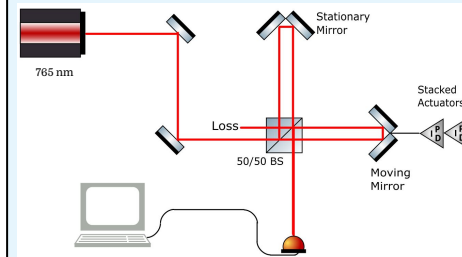
Fourier transform of the interference equation:

$$S(f) = |A_0|^2 \sigma^2 \pi e^{-2\pi^2 \sigma^2 (f_0 - f)^2}$$

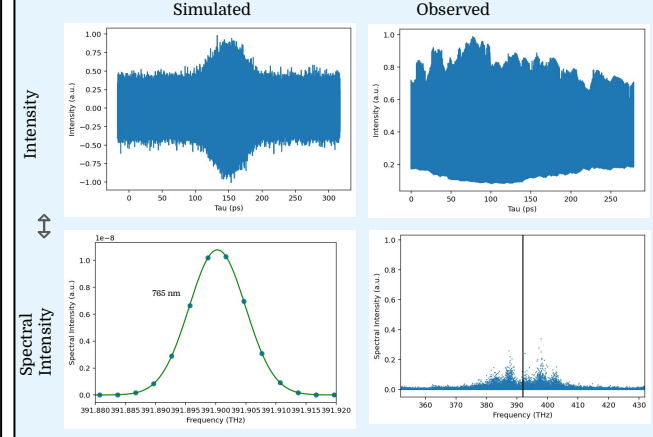
Continuous Wave



Experimental Setup



Pulse



Conclusions

The CW run of the experiment demonstrates issues that will also be present in the pulse. The graph of the CW spectral intensity is centered at the expected frequency, but the unexpected bimodal shape is likely an artifact of the stages shifting over the course of the experiment. In addition, the envelope and apparent beat frequencies of the CW interferogram likely result from the polychromatic nature of the laser and/or alignment issues.

The Fourier transform graph of the pulse has a similar bimodal shape (zoomed in) we expect from the simulation. However, there is a peak at the center of the graph, which could be the frequencies that form the pulse. More analysis is needed to determine if it is so.

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